

The Mars Rover Example

We use a simplified example inspired by the Mars Rover to formalize the reinforcement learning problem.

The Setup

- **States (s):** There are 6 possible positions (boxes) for the rover.
 - $s_1, s_2, s_3, s_4, s_5, s_6$.
 - The rover starts at a specific state (e.g., s_4).
 - **Rewards ($R(s)$):** Each state has an associated reward value.
 - s_1 : **+100** (Highly interesting science target).
 - s_6 : **+40** (Moderately interesting science target).
 - s_2, s_3, s_4, s_5 : **0** (No significant science value).
 - **Terminal States:** Once the rover reaches state 1 or state 6, the "day" ends. It collects the reward and stops moving.
 - **Actions (a):** At each step, the rover can move:
 - **Left** ($\leftarrow$)
 - **Right** ($\rightarrow$)
-

The Reinforcement Learning Loop

At every time step, the following sequence occurs:

1. **Current State (s):** The robot is in a specific position (e.g., state 4).
2. **Reward ($R(s)$):** The robot receives the reward associated with the *current* state (e.g., 0).
3. **Action (a):** The robot chooses an action (e.g., "Go Left").
4. **Next State (s'):** As a result of the action, the robot transitions to a new state (e.g., state 3).

This tuple $(s, a, R(s), s')$ —Current State, Action, Reward, Next State—is the fundamental building block of RL algorithms.

Example Trajectories

- **Option A (Go Left):**
 - Start at 4 ($R = 0$) $\rightarrow$ Go Left $\rightarrow$ At 3 ($R = 0$) $\rightarrow$ Go Left $\rightarrow$ At 2 ($R = 0$) $\rightarrow$ Go Left $\rightarrow$ At 1 ($R = 100$).
 - **Total Reward:** 100. (Plus it reached the best target).

- **Option B (Go Right):**

- Start at 4 ($R = 0$) $\rightarrow$ Go Right $\rightarrow$ At 5 ($R = 0$) $\rightarrow$ Go Right $\rightarrow$ At 6 ($R = 40$).
- **Total Reward:** 40. (Reached a target, but a less valuable one).

- **Option C (Wasted Motion):**

- Start at 4 $\rightarrow$ Go Right (to 5) $\rightarrow$ Go Left (back to 4) $\rightarrow$ Go Left (to 3) ... $\rightarrow$ At 1 ($R = 100$).
- **Result:** Still gets +100 reward eventually, but wasted time and fuel.

The goal of the algorithm is to figure out that **Option A** is the best strategy.